Solubility

1. Correct Answer: (d) All of the above

Explanation:

Gas solubility in liquids depends on:

- **Nature of gas and solvent** (some gases dissolve better)
- **Temperature** (generally decreases with increase in temperature)
- **Pressure** (increases with pressure according to Henry's law)

2. Correct Answer: (d) Solubility of NaCl in it is more than H_2O

Explanation:

Heavy water properties:

- Boiling point **higher** than water
- Chemical reactions **slower**
- **Viscosity higher**
- **Solubility of salts like NaCl is slightly less**, not more

3. Correct Answer: (c) Henry's Law

4. Correct Answer: (a) and (b)

But since the options given are fixed, the **closest correct answer is (a) and (b)** only.

5. Correct Answer: (d) Henry's law

6. Correct Answer: (c) Free molecules

Explanation:

Sugar is a **non-electrolyte**, so it dissolves as **molecules**, not ions.